McGill University
Department of Mathematics and Statistics
MATH 782 Advanced Topics in Statistics

Student Name: Wen, Zehai Student ID# 260956530

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Degree Program and Year: PhD 2 Term: Fall 2026 Winter

Topics to be covered: (Outline the Main reading material.)

Topic Title: Asymptotic Statistics

The course aims to provide an introduction to modern tools and topics in asymptotic statistics. It will be split into three main parts: (A) mathematical and probabilistic foundations; (B) classical large-sample theory; (C) an introduction to minimax rates for nonparametric estimation and semiparametric theory.

In part (A), the focus will be on mathematical reminders, stochastic convergence, uniform integrability, laws of large numbers, central limit theorems, triangular arrays, and asymptotic linearity.
Part (B) will focus on studying the asymptotic properties of various classical estimators, such as the empirical distribution function and sample quantiles, maximum likelihood, M-estimation, Z-estimation, with various applications to estimation and testing.
Part (C) will be an introduction to selected modern topics such as minimax upper/lower bounds for nonparametric regression and semiparametric theory, as time permits.

Method of Evaluation: Assignments: 25%. Project: 50 %. Oral presentation: 25%.

Student Signature and Date: 温泽海 2026 October 8

Instructor Name: Mehdi Dagdoug Signature and Date: Dagdoug

For Undergraduate Students - Chief Advisor Approval:

Professor Djivede Kelome Signature and Date:

For Graduate Students – Graduate Program Director Approval

Professor Brent Pym Signature and Date:

Please return form with all 3 signature via email to the Undergraduate Program Coordinator, Angela White at angela.white@mcgill.ca

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